XentriPay

API Documentation

Collections | Payouts | Checkout

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1. Introduction

This document describes the API endpoints available for initiating **Collections**, **Payouts**, and **Chec kout** payments in the XentriPay system. It includes request and response formats, sample payloads, and error handling details.

XentriPay supports multiple payment methods, including **Mobile Money (MoMo)** and **Card payments**, allowing businesses to collect payments from customers or send payouts using different channels depending on their use case.

In addition to direct API-based collections and payouts, XentriPay also provides a **Checkout API**, which enables session-based payments suitable for web and mobile applications, supporting both Mobile Money and Card payment flows.

2. Base URL

XentriPay provides two environments:

Environment	Base URL
Production	https://xentripay.com
Test	https://merchant.test.xentripay.com

Use the test environment while integrating and validating your implementation. Switch to the production URL only when you are ready to go live.

All endpoint paths in this document are the same across both environments. For example, `/api/collections/initiate` maps to:

- Production: `https://xentripay.com/api/collections/initiate`
- Test: `https://merchant.test.xentripay.com/api/collections/initiate`

3. Authentication

All endpoints require a valid API key passed in the `X-XENTRIPAY-KEY` header.

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Your API key is generated and provided by XentriPay from the merchant dashboard. Keep it secure and do not expose it in client-side or front-end code.

Unauthorized Response

If the API key is missing or invalid, the API returns:

```
{
  "status": 401,
  "message": "Invalid or disabled API key"
}
```

4. Collections API

4.1 Overview and Status Lifecycle

The Collections API allows a business to initiate a payment collection request from a customer. XentriPay supports **Mobile Money (MoMo)** and **Card** as collection payment methods. The payment method is specified using the `pmethod` field in the request body.

> *Minimum amount: 100 RWF per collection.*

How a collection is processed:

1. You call `POST /api/collections/initiate`: the collection is saved with status `PENDING`.
2. For MoMo: the customer receives a prompt on their phone and confirms or declines.
3. For Card: the customer is redirected to the card payment page via the `url` in the response.
4. XentriPay's background scheduler polls the payment gateway and updates the status automatically.

Status	Meaning
PENDING	Payment has been initiated and is awaiting confirmation from the provider.
SUCCESS	Payment was confirmed. The business wallet has been credited.
FAILED	Payment was declined, cancelled, or timed out.

4.2 Initiate a MoMo Collection

Endpoint

```
POST /api/collections/initiate
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "email": "customer@example.com",
  "cname": "John Doe",
  "amount": 500,
  "cnumber": "0780371519",
  "msisdn": "250780371519",
  "currency": "RWF",
  "pmethod": "momo",
  "chargesIncluded": true
}
```

> *Note: amount must be a whole number (no decimals) when currency is RWF. Minimum is 100 RWF.*

Sample Response: HTTP 200 OK


```
{
  "reply": "Thank you for initiating a new Payment. Check your pending transactions on 182*7*1# to
  "url": null,
  "success": 1,
  "authkey": "vzimHGg2_iXcmwpwxdwy_KKW5zMr3_iV",
  "tid": "680780365824",
  "refid": "RefE1108D491A1C",
  "retcode": 0
}
```


> *Note: For MoMo collections, url is null. A URL is only returned for card payments.*

4.3 Initiate a Card Collection

Endpoint

```
POST /api/collections/initiate
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "email": "customer@example.com",
  "cname": "John Doe",
  "cnumber": "0780371519",
  "amount": 500,
  "msisdn": "250780371519",
  "currency": "RWF",
  "pmethod": "cc",
  "redirecturl": "https://yourdomain.com/payment/callback",
  "returl": "https://yourdomain.com/payment/return",
  "chargesIncluded": true
}
```

Sample Response: HTTP 200 OK

```
{
  "reply": "Thank you for initiating a new Payment. Continue to process Card Payment",
  "url": "https://urubutopay.rw/pay-now/card/process/11202512191833068307",
  "success": 1,
  "authkey": "p1wYqVAu1cm1UqfuNcROGNRZSRzH4TGq",
  "tid": "261002756096",
  "refid": "Ref8511ED612198",
  "retcode": 0
}
```


> *Note: For card collections, redirect the customer to the url in the response. This opens the card payment page where the customer enters their card details.*

4.4 Customer Reference (customerRef)

To improve transaction traceability and reconciliation, clients are **strongly encouraged** to include a `customerRef` when initiating a collection.

The `customerRef` is a client-defined identifier that is stored with the transaction. It does not change the API response or payment flow, but allows both you and XentriPay to easily identify, track, and support specific transactions.

Sample Request with customerRef

```
{
  "email": "customer@example.com",
  "cname": "John Doe",
  "cnumber": "0780371519",
  "amount": 500,
  "msisdn": "250780371519",
  "currency": "RWF",
  "pmethod": "momo",
  "customerRef": "ORDER-20240101-001",
  "chargesIncluded": true
}
```

4.5 Check Collection Status

This endpoint returns the current status of a collection. You can use either the `customerRef` you provided at initiation, or the `refid` returned in the initiation response.

Endpoint

```
GET /api/collections/status/{reference}
```

The `{reference}` value can be either the `customerRef` or the `refid` of the collection.

Sample Request

```
GET  https://xentripay.com/api/collections/status/ORDER-20240101-001
```

or

```
GET https://xentripay.com/api/collections/status/RefE1108D491A1C
```

Sample Response

```
{
  "customerRef": "ORDER-20240101-001",
  "rid": "RefE1108D491A1C",
  "status": "SUCCESS",
  "updatedAt": "2025-09-25T23:57:50.672947"
}
```

4.6 Collections Field Descriptions

Request Fields

Field	Type	Required	Description
email	String	Yes	Customer email address.
cname	String	Yes	Customer full name.
amount	Number	Yes	Amount to collect. Must be a whole number for RWF. Minimum: 100.
cnumber	String	Yes	Customer phone number in local 10-digit format (e.g., 0780371519). No spaces or letters.
msisdn	String	Yes	Customer phone number in international format with country code (e.g., 250780371519).
currency	String	Yes	Currency code. Currently only RWF is supported.
pmethod	String	Yes	Payment method. Use momo for Mobile Money or cc for card payment.
chargesIncluded	Boolean	No	If true, the transaction fee is deducted from the specified amount. If false or omitted, the fee is added on top.
customerRef	String	No	Client-defined unique reference for tracking and reconciliation. Strongly recommended.
redirecturl	String	Required for cc	URL to redirect the customer after the card payment is processed (callback URL).
returl	String	Required for cc	URL to return the customer to after the full payment flow completes.
details	String	No	Optional description or note about the payment.

Response Fields

Field	Type	Description
reply	String	Human-readable message from the gateway.
url	String	Card payment URL. Populated for card (cc) payments only. null for MoMo.
success	Number	1 for success, 0 for failure.
authkey	String	Authentication key for this transaction instance.
tid	String	Transaction ID assigned by the payment gateway.
refid	String	XentriPay reference ID. Use this (or your customerRef) to check the transaction status.
retcode	Number	Return code. 0 means the request was accepted. See error codes below.

4.7 Error Responses

HTTP 400: Validation Errors

Scenario	Response
Missing email	{"message": "Email is required", "status": 400}
Invalid email format	{"message": "Email must be a valid email address", "status": 400}
Invalid cnumber format	{"message": "Customer number must be exactly 10 digits with no spaces or letters", "status": 400}
Unsupported currency	{"message": "Currency must be 'RWF'", "status": 400}
Amount too low	{"message": "Amount must be at least 100 RWF", "status": 400}

retcode Error Values

retcode	Meaning
0	Request accepted successfully.
606	Payment failed. Check that the phone number is correct and has sufficient funds.
607	Transaction declined by the mobile money provider.
608	Payment timed out. Please try again.

5. Payouts API

5.1 Overview

The Payouts API allows a business to send money to a recipient via Mobile Money or bank transfer. Payouts are processed through supported providers.

This API is typically used for disbursements such as salaries, refunds, commissions, or partner payments.

> *Note: Payout requests require OTP (One-Time Password) confirmation before being processed. After calling the initiate endpoint, an OTP is sent to the authorized business user. The payout is only processed after OTP confirmation.*

5.2 Initiate a Payout

Endpoint

```
POST /api/payment-requests
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "customerReference": "TXN-20240402-001",
  "telecomProviderId": "63510",
  "msisdn": "0788302208",
  "name": "John Doe",
  "transactionType": "PAYOUT",
  "currency": "RWF",
  "amount": 500
}
```

> *Note: For payouts, msisdn is in local format (e.g., 0788302208) without the country code prefix.*

Sample Response: HTTP 200 OK

```
{
  "id": 2812,
  "businessName": "Acme Business Ltd",
  "customerReference": "TXN-20240402-001",
  "telecomProvider": "MTN MOBILE MONEY",
  "telecomProviderId": "63510",
  "msisdn": "0788302208",
  "transactionType": "PAYOUT",
  "currency": "RWF",
  "amount": 500.00,
  "txnCharge": 10.00,
  "status": "PENDING",
  "statusMessage": "Payment request submitted to eKash successfully, awaiting confirmation.",
  "internalRef": "Ref42FCC70F46DD",
  "remoteIp": "60.108.66.235",
  "paymentChanel": "EKASH",
  "validatedAccountName": "John Doe",
  "externalTransactionRef": null,
  "createdAt": "2025-07-20T06:51:44.404271908",
  "updatedAt": "2025-07-20T06:51:44.404274362"
}
```


Note: After receiving this response, the payout will be held as PENDING until the authorized user confirms via OTP. The OTP is sent automatically to the registered email/phone of the business.

5.3 Check Payout Status

This endpoint returns the current status of a payout from the payment provider. Use the `customerReference` you provided when initiating the payout.

Important: The status returned here reflects the status from the external payment provider, not the internal XentriPay system status. See Section 5.4 for the mapping between the two.

Endpoint

```
GET /api/payment-requests/check-status?customerRef={customerReference}
```

Sample Request

```
GET https://xentripay.com/api/payment-requests/check-status?customerRef=TXN-20240402-001
```

Sample Response

```
{
  "timestamp": "2025-07-22 10:42:52",
  "message": "Success",
  "data": {
    "status": "COMPLETED",
    "reference_number": "00461025533065",
    "amount": "500.0"
  }
}
```

5.4 Payout Status Flow

```
PENDING - COMPLETED (provider) - SUCCESSFUL (internal)
PENDING - FAILED
PENDING - REVERSED
```

Status	Meaning
PENDING	Payout submitted, awaiting provider confirmation.
COMPLETED	Provider has fully processed and finalized the payout.
SUCCESSFUL	XentriPay's internal confirmation that payout is finalized.
FAILED	Payout was declined or failed at the provider level.
REVERSED	Payout was reversed after initial processing.

Note: COMPLETED is only used for payouts. Collections never use this status: they use PENDING, SUCCESS, or FAILED.

5.5 Payouts Field Descriptions

Request Fields

Field	Type	Required	Description
customerReference	String	Yes	Unique reference ID for this payout, defined by you. Must be unique per transaction.
telecomProviderId	String	Yes	ID of the payment provider. See Section 6 for the full list.
msisdn	String	Yes	Recipient phone number in local format (e.g., 0788302208). Do not include the country code.
name	String	Yes	Full name of the payout recipient.

Field	Type	Required	Description
transactionType	String	Yes	Must be "PAYOUT".
currency	String	Yes	Currency code. Currently only RWF is supported.
amount	Number	Yes	Amount to pay out before charges.

Response Fields

Field	Type	Description
id	Number	XentriPay internal record ID.
businessName	String	Name of the business initiating the payout.
customerReference	String	The reference provided in the request.
telecomProvider	String	Human-readable provider name (e.g., "MTN MOBILE MONEY").
telecomProviderId	String	ID of the provider used.
msisdn	String	Recipient phone number.
transactionType	String	Transaction type (e.g., "PAYOUT").
currency	String	Currency code.
amount	Number	Amount paid out before charges.
txnCharge	Number	Transaction fee applied.
status	String	Current status (PENDING, SUCCESSFUL, FAILED, REVERSED).
statusMessage	String	Gateway message describing the result.
internalRef	String	XentriPay internal reference for this transaction.
externalTransactionRef	String	Reference from the payment gateway. May be null initially.
paymentChanel	String	Payment gateway used (e.g., "EKASH").
validatedAccountName	String	Account name returned by the provider after validation.
remoteIp	String	IP address from which the request was made.
createdAt	DateTime	Timestamp when the payout was created.
updatedAt	DateTime	Timestamp when the payout was last updated.

6. Bank and Mobile Money Provider IDs

Use the appropriate `telecomProviderId` from the table below when initiating payouts.

Provider Name	ID
MTN MOBILE MONEY	63510
AIRTEL RWANDA	63514

Provider Name	ID
SPENN	63509
BANQUE DE KIGALI	040
BANQUE POPULAIRE DU RWANDA	400
EQUITY BANK	192
ECOBANK RWANDA	100
ACCESS BANK RWANDA	115
GUARANTY TRUST BANK (RWANDA)	070
INVESTMENT AND MORTGAGE BANK	010
NATIONAL COMMERCIAL BANK OF AFRICA	025
URWEGO OPPORTUNITY BANK	145
ZIGAMA CREDIT AND SAVINGS SCHEME	800
BANK OF AFRICA RWANDA	900
UNGUKA BANK	950
BANQUE NATIONALE DU RWANDA	951

7. Checkout API

7.1 Overview and Flow

The Checkout API provides a **session-based payment flow**. Unlike direct collections, it separates the session creation from the actual payment step, making it suitable for hosted payment pages and e-commerce flows.

Two-step flow:

- **Step 1: Create a session:** Call `POST /api/checkout/sessions` with the amount and redirect URL.

You receive a session ID and a hosted checkout URL.

- **Step 2: Process payment:** Either redirect the customer to the `checkoutUrl` (hosted UI), or call `POST /api/checkout/sessions/{sessionId}/pay` directly with the customer's payment details.

> *Important: The amount is fixed at session creation and cannot be changed in the payment step.*

7.2 Step 1: Create a Checkout Session

This step creates a payment session with a fixed amount. Use the same endpoint for both MoMo and card sessions: the payment method is chosen in Step 2.

Endpoint

```
POST /api/checkout/sessions
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "amount": 200,
  "customerFinalUrl": "https://yourdomain.com/payment/result",
  "currency": "RWF"
}
```

Response

```
{
  "id": "cs_69070fe2-b9c7-4c42-9e01-6a73a1ead2a1",
  "checkoutUrl": "https://xentripay.com/checkout/cs_69070fe2-b9c7-4c42-9e01-6a73a1ead2a1",
  "status": "CREATED"
}
```

Use the `id` as the `sessionId` in Step 2. You may redirect the customer to `checkoutUrl` to let XentriPay's hosted page handle the payment, or proceed directly to Step 2.

7.3 Step 2: Pay with Mobile Money

Endpoint

```
POST /api/checkout/sessions/{sessionId}/pay
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "customerRef": "CHK-081025132217-02",
  "email": "customer@example.com",
  "cname": "John Doe",
  "cnumber": "0780371519",
  "msisdn": "250780371519",
  "details": "Payment for Order #1234",
  "currency": "RWF",
  "pmethod": "momo",
  "chargesIncluded": true
}
```

Response

```
{
  "status": "PENDING",
  "redirectTo": "https://yourdomain.com/payment/result?session_id=cs_69070fe2-b9c7-4c42-9e01-6a73a
  "gatewayUrl": null,
  "paymentMethod": "momo"
}
```

Note: For MoMo payments, gatewayUrl is null. The customer completes the payment on their phone after receiving a prompt.

7.4 Step 2: Pay with Card

Endpoint

```
POST /api/checkout/sessions/{sessionId}/pay
```

Request Headers

```
X-XENTRIPAY-KEY: <your_api_key>
Content-Type: application/json
```

Request Body

```
{
  "customerRef": "CHK-081025132217-03",
  "email": "customer@example.com",
  "cname": "John Doe",
  "cnumber": "0780371519",
  "msisdn": "250780371519",
  "details": "Payment for Order #1234",
  "currency": "RWF",
  "pmethod": "cc",
  "gatewayRedirectUrl": "https://yourdomain.com/payment/callback"
}
```

Response

```
{
  "status": "PENDING",
  "redirectTo": "https://yourdomain.com/payment/result?session_id=cs_0ef17390-8a76-4fff-993d-722e1
  "gatewayUrl": "https://urubutopay.rw/pay-now/card/process/11202512191313196629", "paymentMethod"
  "cc"
}
```

Note: For card payments, redirect the customer to gatewayUrl. This opens the card payment page (powered by Urubuto) where the customer enters their card details. gatewayRedirectUrl is required for card payments and tells Urubuto where to redirect after processing.

7.5 Check Checkout Session Status

Returns the current status of a checkout session using the `refid` from the session's collection record.

Endpoint

```
GET /api/checkout/sessions/status/{refid}
```

Sample Request

```
GET https://xentripay.com/api/checkout/sessions/status/RefE1108D491A1C
```

Sample Response

```
{
  "refid": "RefE1108D491A1C",
  "status": "SUCCESS"
}
```

7.6 Checkout Field Descriptions

Session Creation Fields (Step 1)

Field	Type	Required	Description
amount	Integer	Yes	Amount to charge. Must be a whole number. Fixed for the life of the session.
customerFinalUrl	String	Yes	URL to redirect the customer once the payment flow completes.
currency	String	No	Currency code. Defaults to RWF. Currently only RWF is supported.

Payment Fields (Step 2)

Field	Type	Required	Description
customerRef	String	No	Client-defined reference for tracking this payment. Recommended.
email	String	No	Customer email address.
cname	String	No	Customer full name.
cnumber	String	No	Customer phone number in local 10-digit format.
msisdn	String	No	Customer phone number in international format (e.g., 250780371519).
details	String	No	Description of what the payment is for.
currency	String	No	Currency code. Defaults to session currency if not provided.
pmethod	String	Yes	Payment method: momo for Mobile Money, cc for card.
chargesIncluded	Boolean	No	If true, fee is deducted from the amount. Defaults to session setting.
gatewayRedirectUrl	String	Required for cc	URL to redirect after the card gateway processes the payment. Not required for MoMo.

Session Creation Response Fields

Field	Type	Description
id	String	Unique session ID. Use as sessionId in the pay endpoint.
checkoutUrl	String	Hosted checkout page URL. Redirect the customer here to let XentriPay handle the payment UI.
status	String	Session status. Always "CREATED" when the session is first created.

Payment Response Fields

Field	Type	Description
status	String	Payment status. PENDING immediately after initiation.
redirectTo	String	URL to redirect the customer after payment is initiated (includes session ID and status as query params).
gatewayUrl	String	Card payment page URL. Populated only for card (cc) payments. null for MoMo.
paymentMethod	String	Payment method used (momo or cc).

7.7 Important Notes

- The `amount` is fixed at session creation and cannot be changed in the payment step.

- `gatewayRedirectUrl` is **required** for card payments and **not required** for MoMo payments.
- Each `customerRef` should be unique to ensure accurate tracking and reconciliation.
- For MoMo: `gatewayUrl` in the response will be `null`. The customer completes payment on their phone.
- For card: redirect the customer to `gatewayUrl` to complete payment on the Urubuto card page.
- The `customerFinalUrl` (set at session creation) is where the customer is sent after the payment completes, regardless of the payment method.

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